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HALF YEARLY EXAMINATION - 2025

CLASS:XII **BIOLOGY** **Reg.No**

Time : 3.00 Hours **BIO-BOTANY** **MAX MARKS : 70**

MARKS : 35

Instructions : Candidate should answer Part - I (Bio-Botany) & Part - II (Bio-Zoology) in Separate answer books.

SECTION - I

Note : i) Answer all the questions. **8X1=8**

ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. Transmitting tissue is found in
   a) Integument b) Micropylar region of ovule c) Pollen tube wall d) Stylar region of gynoecium
2. Among the following characters which one was not considered by Mendel in his experimentation on Pea.
   a) Stem - Tall or dwarf b) Trichomal glandular or non-glandular
   c) Seed coat - Green or yellow d) Pod Shape - Inflated or constricted
3. In which of the following double monosomics are observed?
   a) Wheat b) Rye c) Maize d) Oenothera
4. The Process of recombinant DNA technology has the following steps. Pick out the correct sequence of them.
   (i) Amplication of the gene. (ii) Insertion of recombinant DNA into the host cells.
   (iii) Cutting of DNA at specific location using restriction enzyme.
   (iv) Isolation of genetic material (DNA)
   a) (ii), (iii), (iv), (i) b) (iv), (ii), (iii), (i) c) (i), (ii), (iii), (iv) d) (iv), (iii), (i), (ii)
5. Match the following and choose the correct combination from the options given below.

Column - I (Interactions)		Column - II (Examples)
(i) Mutalism	-	(I) Trichoderma and Penicillium
(ii) Commensalism	-	(ii) Balanophora and Orobanche
(iii) Parasitism	-	(iii) Orchids and Ferns
(iv) Predation	-	(iv) Lichen and Mycorrhiza
(v) Amensalism	-	(v) Nepenthes and Diaonaea

	(i)	(ii)	(iii)	(iv)	(v)
a)	(i)	(ii)	(iii)	(iv)	(v)
b)	(ii)	(iii)	(iv)	(v)	(i)
c)	(iii)	(iv)	(v)	(i)	(ii)
d)	(iv)	(iii)	(ii)	(v)	(i)

6. Who is known as (named as) Forest Man of India?
   a) Sudhir Kumar Sopory b) Sundarlal Bahuguna c) Panduranga Hegde d) Jadav Molai Payeng
7. Which of the following is incorrectly paired?
   a) Wheat - Himgiri
   b) Milch breed - Sahiwal
   c) Rice - Ratna
   d) Pusa Komal - Brassica (cow pea)
8. Assertion : Tumeric fights various kinds of cancer.
   Reason : Curcumin is an anti-oxidant present in turmeric.
   a) Assertion is correct, Reason is wrong b) Assertion is wrong, Reason is correct
   c) Both are correct d) Both are wrong

SECTION - II

II Answer any Four questions: **4X2=8**

9. What is called double fertilization?
10. What is known as atavism?
11. What is called Single Cell Protein?
12. Green algae are not likely to be found in the deepest strata of the ocean. Give at least one reason.
13. Pyramid of energy is always upright. Give reasons.
14. What is Ozone hole?

SECTION - III

III Answer any Three of the following. Q.No.19 is compulsory **3X3=9**

15. List out the functions of Tapeatum.
16. What is called Cytoplasmic inheritance?
17. Draw any six types of aneuploidy.
18. What is called Albedo effect? Write its effects.
19. Write the objectives of plant breeding.

SECTION - IV

IV Answer all the questions. **2X5=10**

20. (a) What is called Endosperm? Describe its types. **(OR)**
    (b) Write the application of Biotechnology.
21. (a) If we fail to protect environment, we will fail to save posterity. To justify this, What we should have practice in our day today life? **(OR)**
    (b) Which TSM is widely practised and culturally accepted in Tamil Nadu? Explain it.

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BIO-ZOOLOGY

MARKS : 35

SECTION - I

8x1=8

Note : (i) Answer all the questions.
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. The mode of sexual reproduction in bacteria is by
   (a) Formation of gamete (b) Endospore formation (c) Conjugation (d) Zoospore formation
2. Which female gland performs a function similar to the male bulbourethral gland?
   (a) Skene's gland b) Mammary gland c) Bartholin's gland (d) Thyroid gland
3. Select the incorrect action of hormonal contraceptive pills from the following
   (a) Inhibition of spermatogenesis. (b) Inhibition of ovulation.
   (c) Changes in cervical mucus impairing its ability to allow passage and transport of sperms.
   (d) Alteration in uterine endometrium to make it unsuitable for implantation.
4. What is the basis for the difference in the synthesis of the leading and lagging strand of DNA molecules?
   (a) Origin of replication occurs only at the 5' end of the molecules.
   (b) DNA ligase works only in the 3' → 5' direction.
   (c) DNA polymerase can join new nucleotides only to the 3' end of the growing stand.
   (d) Helicases and single-strand binding proteins that work at the 5' end.
5. The infective stage of Entamoeba histolytica is:
   (a) Merozoite (b) Trophozoite (c) Oocyst (d) Sporozoite
6. Match the microbe in List A with the organic acid it produces in List B.

List A (Microbe)	List B (Organic Acid)	
1. Aspergillus niger	A. Acetic acid	(a) 1-A, 2-D, 3-C, 4-B
2. Acetobacter aceti	B. Lactic acid	(b) 1-D, 2-A, 3-B, 4-C
3. Lactobacillus	C. Butyric acid	(c) 1-D, 2-C, 3-A, 4-B
4. Clostridium butyricum	D. Citric acid	(d) 1-A, 2-B, 3-D, 4-C

7. Vaccines that use components of a pathogenic organism rather than the whole organism are called
   a) Subunit recombinant vaccines b) Attenuated recombinant vaccines
   c) DNA vaccines d) Conventional vaccines
8. Which Air Quality index (AQI) range is considered "Unhealthy for Sensitive Groups"?
   a) 0 - 50 b) 51 - 100 c) 101 - 150 d) 151 - 200

SECTION - II

II Answer any 4 questions. 4X2=8

9. What are vestigial organs? Name any two vestigial organs present in human beings.
10. Write short note on the Ideonella sakaiensis.
11. What are the major functions of soil?
12. Define - Co Extinction.
13. What is parthenogenesis? Give two examples from animals.
14. Mention the importance of the position of the testes in humans.

SECTION - III

III Answer any 3 questions. Question No.19 is compulsory. 3x3=9

15. Amniocentesis, the foetal sex determination test, is banned in our country, Is it necessary? comment.
16. Differentiate - Anadromous migration and Catadromous migration.
17. Mention the advantages of cloning.
18. What steps should be taken to prevent Erythroblastosis foetalis?
19. UV treatment is better than chlorination to disinfect the wastewater.Why?

SECTION - IV

IV Answer all the questions. 2X5=10

20. a) Describe the structure of the human ovum with a neat labelled diagram. **(OR)**
    b) Explain the structure of immunoglobulin with suitable diagram.
21. a) What are the reasons for the richness of biodiversity in the Tropics. **(OR)**
    b) Explain the formation of a nucleosome.

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HALF YEARLY EXAMINATION – 2025 – KRISHNAGIRI DISTRICT

CLASS: 12 BIO - BOTANY

SECTION – I

1. d) Stylar region of gynoecium
2. b) Trichomal glandular or non-glandular
3. c) maize
4. d) (iv), (iii), (i), (ii)
5. d) (i)-IV, (ii)-III, (iii)-II, (iv)-V, (v)-I
6. d) Jadav Molai Payeng
7. d) Pusa Komal - Brassica
8. c) Both are correct

SECTION - II

II Answer any Four questions: (4 x 2 = 8)

9. What is called double fertilization?

Both the male gametes released from a male gametophyte are involved in the fertilization. They fertilize two different components of the embryo sac.
Since both the male gametes are involved in fertilization, the phenomenon is called double fertilization

10. What is known as atavism?

Atavism is a modification of a biological structure whereby an ancestral trait reappears after having been lost through reemergence of sexual reproduction in the flowering plant Hieracium pilosella is the best example for Atavism in plants.

11. What is called Single Cell Protein?

Single cell proteins are dried cells of microorganism that are used as protein supplement in human foods or animal feeds

12. Green algae are not likely to be found in the deepest strata of the ocean. Give at least one reason.

1.The deeper region of a ocean below the limnetic zone is called profundal zone.

2.In this zone there is no effective sunlight. Hence, green algae are not found in the deepest strata of the ocean.

13. Pyramid of energy is always upright. Give reasons.

1.In the energy pyramid producers to tropical level the energy level is gradually decreasing.
2.So, energy pyramis is always upright

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14. What is Ozone hole?

1.The depletion of ozone layer over restricted area is called ozone hole.

2.This ozone shield is being damaged by CFC, SO_2, NO_2 and CO_2 .

SECTION - III

III Answer any Three of the following. Q.No. 19 is compulsory (3 x 3 = 9)

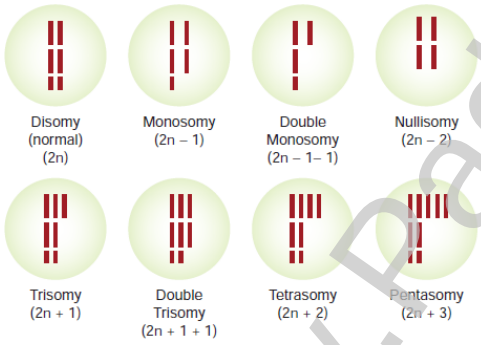
15. List out the functions of Tapetum.

1.It supplies nutrition to the developing microspores.
2.It contributes sporopollenin through ubisch bodies thus plays an important role in pollen wall formation.
3.Exine proteins responsible for 'rejection reaction' of the stigma are present in the cavities of the exine. These proteins are derived from tapetal cells.

16. What is called Cytoplasmic inheritance?

Cytoplasmic organelles such as chloroplast and Mitochondria act as inheritance vectors. It is called as Cytoplasmic inheritance.

17. Draw any six types of aneuploidy.



18. What is called Albedo effect? Write its effects.

Aerosols with small particles is reflecting the solar radiation entering the atmosphere. This is known as Albedo effect or Green House effect.

Effects:

1.It reduce the photosynthesis and respiration rate.
2.It creat acid rain.
3.It destroy the ozone layer.

19. Write the objectives of plant breeding.

- To increase yield, vigour and fertility of the crop
- To increase tolerance to environmental condition, salinity, temperature and drought.
- To prevent the premature falling of buds, fruits etc.
- To improve synchronous maturity.
- To develop resistance to pathogens and pests.
- To develop photosensitive and thermosensitive varieties.

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SECTION - IV

IV Answer all the questions. (2 x 5 = 10)

20. (a) What is called Endosperm? Describe its types. (OR)

(i) Endosperm:-

1.The primary endosperm nucleus (PEN) divides immediately after fertilization but before the zygote starts to divide, into an endosperm.

2.Depending upon the mode of development three types of endosperm are recognized in angiosperms.

(ii) Types of Endosperm:-

1.Nuclear endosperm 2.Cellular endosperm 3.Helobial endosperm

1.Nuclear endosperm:-

(i)Primary Endosperm Nucleus undergoes several mitotic divisions without cell wall formation thus a free Nuclear condition exists in the endosperm.

(ii)Examples: *Coccinia*, *Capsella* and *Arachis.*

2.Cellular endosperm:-

(i)Primary endosperm nucleus divides into 2 nuclei and it is immediately followed by wall formation. (ii)Subsequent divisions also follow cell wall formation.

(iii)Examples: *Adoxa, Helianthus* and *Scoparia.*

3.Helobial endosperm:-

(i)Primary Endosperm Nucleus moves towards base of embryo sac and divides into two nuclei. (ii)Cell wall formation takes place leading to the formation of a large micropylar and small chalazal chamber. (iii)Examples : *Hydrilla* and *Vallisneria*.

(b) Write the application of Biotechnology.

1.Biotechnology is widely used in various fields like agriculture, medicine, environment and commercial industries.
2.It is used in pesticide resistant, stress resistant and disease resistant varieties in agricultural crops.
3.E.coli bacteria is used to produce human insulin for insulin deficiency disorder in human beings.
4.Single cell protein from Spirulina is utilized in food industries.
5.Biotechnology also used in secondary metabolites, biofertilizers, biopesticides and enzymes production.
6.Biochip based biological compter is one of the success of biotechnology.
7.Biomass energy, biofuel and bioremediation techniques are done by the environmental biotechnology.

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21. (a) If we fail to protect environment, we will fail to save posterity. To justify this, What we should have practice in our day today life? (OR)

Therefore, we have to practice the following in our day today life:
- Buy and use only ecofriendly products and recycle them.
- Grow more trees
- Choose sustained farm products (vegetables, fruits, greens, etc.)
- Reduce the use of natural resources.
- Recycle the waste and reduce the amount of waste you produce.
- Reduce consumption of water and electricity.
- Reduce or eliminate the use of house-hold chemicals and pesticides.
- Maintain your cars and vehicles properly. (In order to reduce carbon emission)
- Create awareness and educate about ecosystem protection among your friends and family members

(b) Which TSM is widely practised and culturally accepted in Tamil Nadu? Explain it.

Siddha is the most popular, widely practiced and culturally accepted system in Tamil Nadu.

Siddha system of medicine:

1.It is based on the texts written by 18 Siddhars.
2.Siddha is principally based on Panchabudha philosophy.
3.This system working based on Vatam, Pittam and Kapam.
4.The drug sources of Siddha is Plants and Animal parts.
5.800 different herbs are used as a drug for this system.

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